DO NOW — RECALL

Lesson 061 · Thinking Ahead: Inputs, Preconditions and

Name: _____ Class: _____ Date: _____

1. What is an abstraction assumption?

2. Why validate inputs before processing?

3. What is a reusable component?

4. Array a starts as [2,3,5,7]. Execute a[1]=a[0]+a[2], then a[2]=a[1]-a[3]. Trace the final array, using zero-based indexing.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

A simplifying rule in a model.

2. Model answer

Invalid values can violate preconditions.

3. Model answer

A method/module designed for use in several parts of software.

4. Model answer

[2,7,0,7]. The first assignment sets a[1] to 7; the second uses that updated value, so a[2]=7-7=0.

By the end, you can…

• Specify the inputs and outputs before choosing algorithms: ticket count and unit price determine total charge.

• Make preconditions explicit. Invalid negatives or overflow can produce incorrect outputs; validation and appropriate types reduce risks.

• Caching saves a previously computed result for reuse, trading memory and stale-result risk against repeated calculation time. Reusable methods reduce repeated code.

Vocabulary — know these terms

precondition — Requirement for valid use.

postcondition — Expected condition after execution.

cache — Saved result for quicker reuse.

module — Reusable program component.

NEW LEARNING

Read, annotate and explain

Identify

Specify the inputs and outputs before choosing algorithms: ticket count and unit price determine total charge.

Model

Make preconditions explicit. Invalid negatives or overflow can produce incorrect outputs; validation and appropriate types reduce risks.

Evaluate

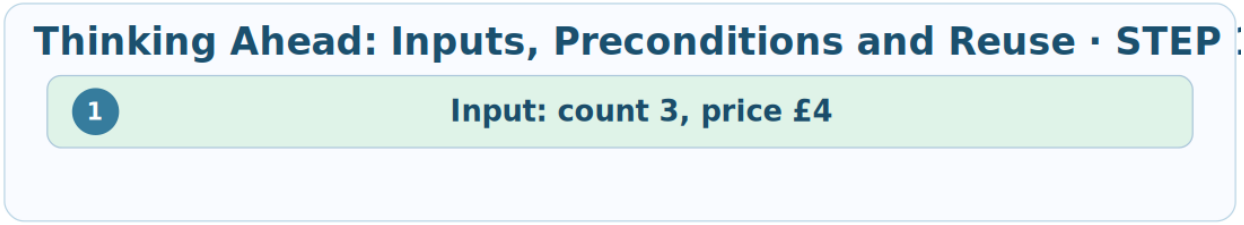
Caching saves a previously computed result for reuse, trading memory and stale-result risk against repeated calculation time. Reusable methods reduce repeated code.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

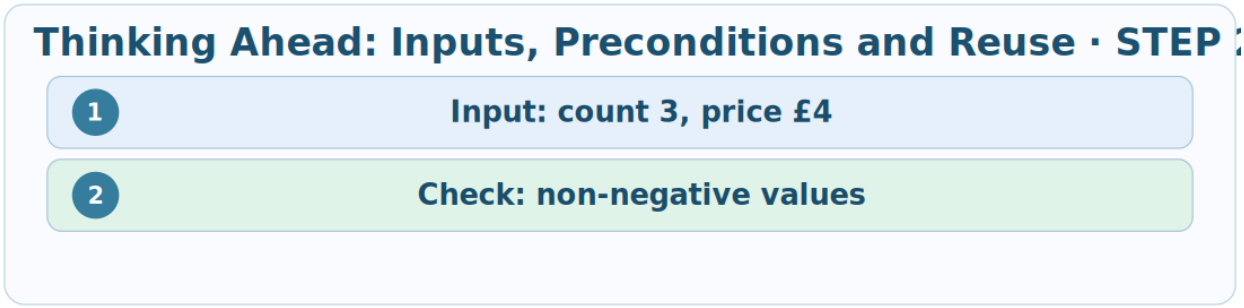
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



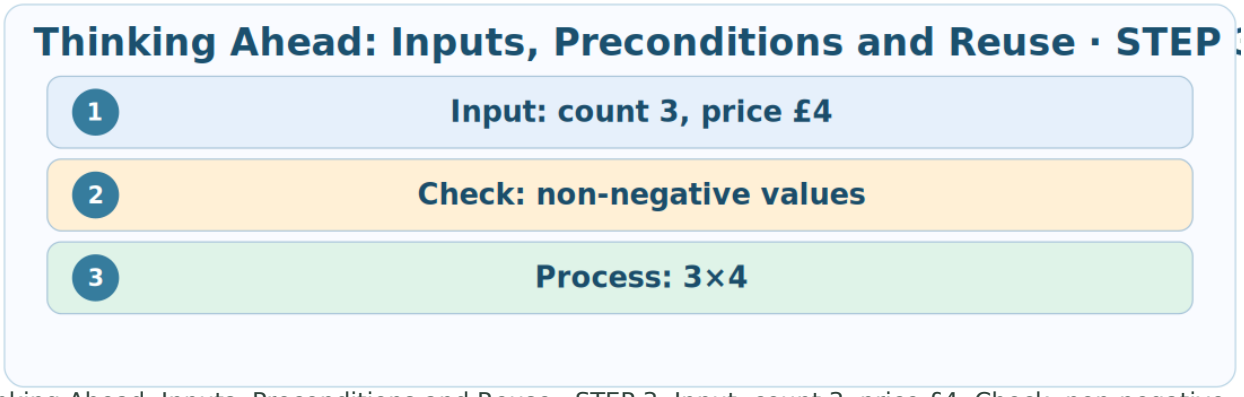
Thinking Ahead: Inputs, Preconditions and Reuse · STEP 1: Input: count 3, price £4

New Learning



Thinking Ahead: Inputs, Preconditions and Reuse · STEP 2: Input: count 3, price £4; Check: non-negative values

New Learning

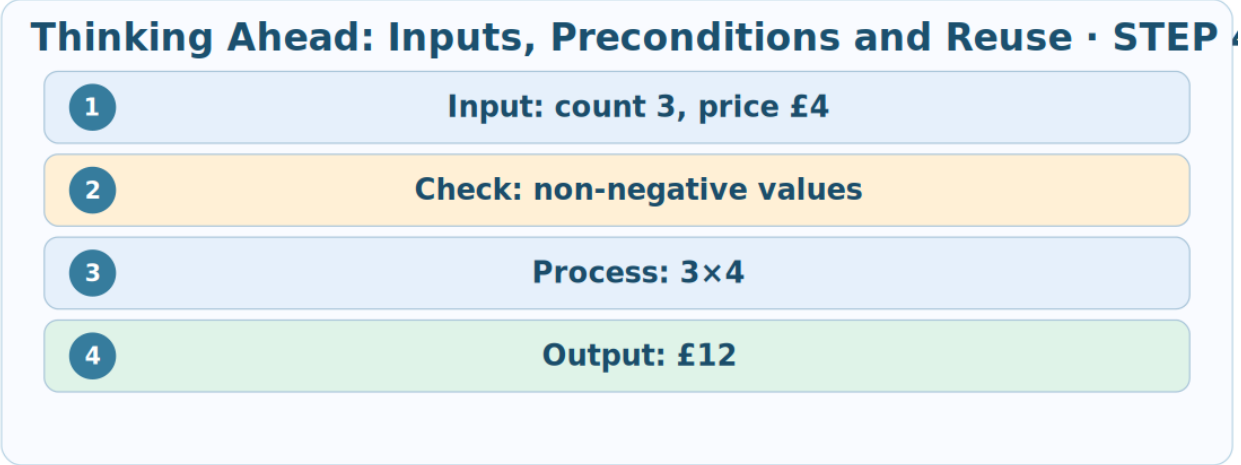


Thinking Ahead: Inputs, Preconditions and Reuse · STEP 3: Input: count 3, price £4; Check: non-negative values; Process: 3×4

New Learning · original lesson diagrams

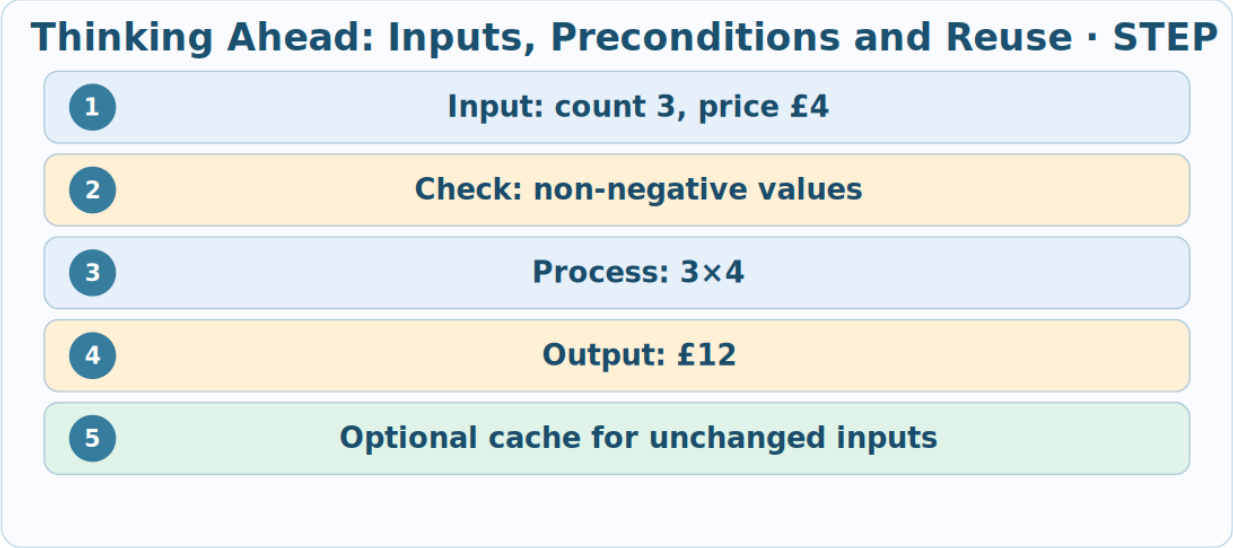
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Thinking Ahead: Inputs, Preconditions and Reuse · STEP 4: Input: count 3, price £4; Check: non-negative values; Process: 3×4; Output: £12

New Learning



Thinking Ahead: Inputs, Preconditions and Reuse · STEP 5: Input: count 3, price £4; Check: non-negative values; Process: 3×4; Output: £12; Optional cache for unchanged inputs

I DO — WORKED EXAMPLE

Follow the worked method

A function doubles an integer. State input/output and a precondition if it accepts values 0–50 only.

Explain the decision

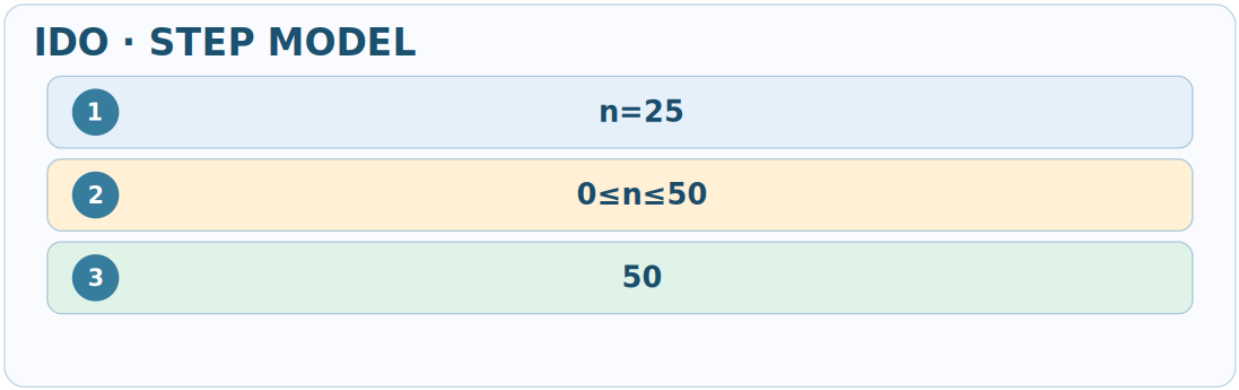
The algorithm's result can be correct mathematically but invalid under the declared application rules.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A function doubles an integer. State input/output and a precondition if it accepts values 0–50 only.

The algorithm's result can be correct mathematically but invalid under the declared application rules.



IDO · STEP MODEL: n=25; 0≤n≤50; 50

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

Guided 1 — TRY

A weather app caches a temperature for 10 minutes. Give a benefit and drawback.
Source code / input:
Avoids frequent costly API calls; can display a stale temperature until cache refresh.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

For function TicketCost(count,price), state two inputs, output and one precondition. [5]

Question 2

Give a scenario where caching a result saves time, plus a stale-data risk and one invalidation strategy. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

Question 3

An API-dependent school timetable app calls same room lookup 200 times. Design a reusable method with cache and describe two tests. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

Caching a result can never affect correctness.

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

Guided 1 — CHECK

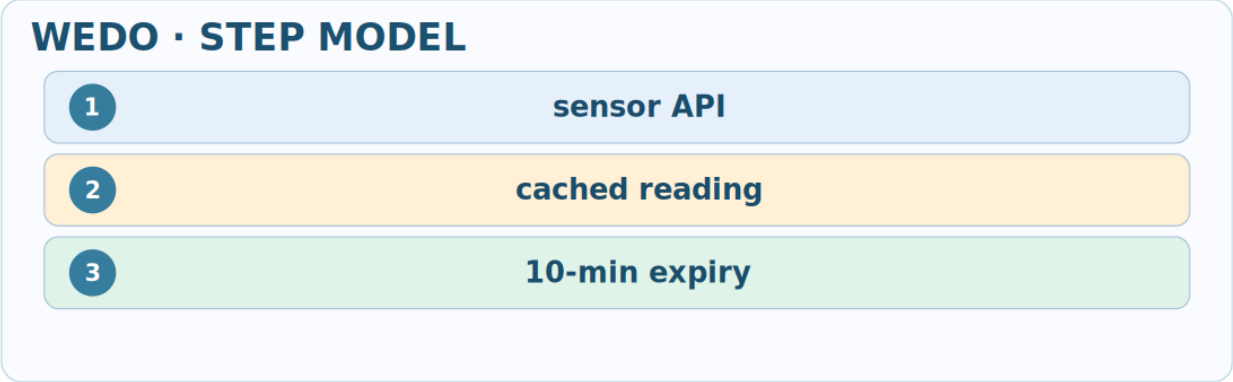
Explain the decision
Explain caching's time/network benefit and correctness trade-off.

We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A weather app caches a temperature for 10 minutes. Give a benefit and drawback.

Explain caching's time/network benefit and correctness trade-off.



WEDO · STEP MODEL: sensor API; cached reading; 10-min expiry